The examination is being carried out on the **following application documents**

Description, Pages

2-30	as originally filed	
1, 1A	filed in electronic form on	26-04-2021

Claims, Numbers

1-3	filed in electronic form on	26-04-2021

Drawings, Sheets

1/5-5/5	as originally filed

Reference is made to the following documents:

D1 JP 2001 255045 A (YAMAHA MOTOR CO LTD) 21 September 2001 (2001-09-21)

D2 JP 2012 017974 A (HITACHI APPLIANCES INC) 26 January 2012 (2012-01-26)

Art. 123(2), 76(1) EPC

Dependent **claim 2** does not fulfill the requirements of Article 76(1) EPC since the parent application does not disclose a **"U-shaped connector"**, in a generalized way, between the **"gas stop valve (5)"** and the **"gas filter (6)"**. In his letter of 26 April 2021, the applicant states that a U-shaped connector is disclosed in Fig. 4, where it is shown as connecting elements with reference numerals 5 and 6. However, Fig. 4 shows a detailed embodiment disclosing a realization of a complete part of the outdoor unit with various mutual connections between individual elements which cannot be structurally and functionally separated from the inverted **"U-shaped"** connection between the **"gas stop valve (5)"** and the **"gas filter (6)"**. The amendment therefore represents an inadmissible intermediate generalization (Guidelines H-V 3.2.1). It is therefore proposed to delete present dependent **claim 2**.

Art. 54 and 56 EPC

The subject-matter of independent **claim 1** is not new (Article 54 EPC) in view of the disclosure of document **D1**. The arguments put forward by the applicant in his letter dated 26 April 2021 do not allay these objections.

With respect to the arguments presented by the applicant in his letter, his attention is drawn to the fact that, under Article 84 EPC, the features of the claims are to be given the broadest meaning given to them by the person skilled in the art. If a feature is to have a more particular meaning based on the description, amendment of the claims is to be carried out (Guidelines F-IV 4.2). In its broadest possibility of interpretation, the person skilled in the art will understand based on his general technical knowledge that a **"flow path that extends in a horizontal direction"** also corresponds to a horizontal direction of a flow path which only forms a sub-part of a larger, overall vertical flow-path. With respect to **D1**, this is the case in view of the horizontal flow across filter element 74c, see Figure 5(B).

Summarizing, therefore, the subject-matter of present independent **claim 1** is not new (Article 54 EPC) in view of the disclosure of document **D1**. Document **D1** (see Figures 1, 3, 5) shows an outdoor unit *(D1: 1a)* suitable for use as part of a heat pump, the outdoor unit comprising: a package that defines a chamber *(D1: engine room 102 and heat exchange chamber 101)*; a liquid stop valve *(D1: stop valve 73)* disposed in the chamber; and a liquid filter *(D1: filter 74)* disposed in the chamber, the liquid filter *(D1: filter 74)* being connected to the liquid stop valve *(D1: stop valve 73)* and configured to be connected, via a liquid pipe *(D1: 48)*, to an indoor unit *(D1: 1b)* of the heat pump, a compressor *(D1: 20)* disposed in the chamber *(D1: implicitly in the engine room 102, where it is connected to engine 2, also see paragraph [0011])*; a gas stop valve *(D1: valve 77, arranged upstream from the strainer 78 and upstream from the joint 47 between the outdoor unit 1a and the gas transfer line 49, see Figure 1)* disposed in the chamber *(D1: see Figure 3)*; and a gas filter *(D1: 78)* disposed in the chamber *(D1: see Figure 3)*, the gas filter *(D1: 78)* being connected to the gas stop valve *(D1: 77)* and configured to be connected, via a gas pipe *(D1: 49)*, to the indoor unit *(D1: 1b)* of the heat pump; wherein the liquid stop valve *(D1: stop valve 73)* has a flow path that extends in an up-down direction *(D1: see Figure 3, flow path from inlet towards outlet arranged overall vertically)* and the liquid filter *(D1: filter 74)* has a flow path that extends in a horizontal direction *(D1: see Figure 5(B) where the flow path inside the filter across filter element 74C is horizontal);* a gas flow path *(D1: 30, 41, 45)* is further defined from the compressor *(D1: 20)* to the gas stop valve *(D1: 77)*, and from the gas stop valve *(D1: 77)* to the gas filter *(D1: 78)*, and the gas filter *(D1: 78)* to the indoor unit *(D1: 1b)*

through the gas pipe *(D1: 49)*, the gas filter *(D1: 78)* is connected to a flange *(D1: 115, 117)* which is disposed on the gas pipe *(D1: upstream of valve 79 forming part of said gas pipe)* configured to be connected to the indoor unit *(D1: 1b)*.

In order to arrive at a subject-matter that may fulfill the requirements of the EPC with respect of novelty and inventive step (Articles 54 and 56 EPC) with respect to the closest prior art shown by document D1 and in further view of **D2, it is proposed to** include the subject-matter of dependent **claim 3** and the additional optional feature of **"overlapping height range[s]" of paragraph [0012] of the originally filed description** in the subject-matter of independent **claim 1**. These features would solve the problem of improved mountability and reduced space requirements and thereby fulfill the advantages addressed by the applicant in his letter of 26 April 2021.

Of course, the applicant may also amend the independent claim differently, as long as the resulting subject-matter represents an inventive contribution over the cited prior art and complies with the further requirements of the EPC. When amending the application, the applicant should ensure that the requirements of Article 123(2) EPC are fulfilled, particularly avoiding inadmissible intermediate generalizations (see the Guidelines H-V 3.2.1).

FURTHER COMMENTS

The subject-matter of dependent **claim 3** is also not new (Article 54 EPC) in view of document **D1** (see Figures 3, 5), which also shows an overall up-down, i.e. overall vertical, orientation of both filter and valve.

Notwithstanding its failure to comply with the requirements of Article 123(2) EPC, see above, the subject-matter of dependent **claim 2** is obvious (Article 56 EPC) in view of the closest prior art shown by document **D1**. The skilled person is aware of the possibility of connecting the gas stop valve and the gas filter to each other via a U-shaped connector from his **general technical knowledge** for compensating thermal expansion, and will apply this to the configuration of document **D1** without inventive skill if the circumstances make this desirable.

When submitting an amended set of claims, the applicant should take into account all of the objections raised above and further the following points:

1 The independent claims should be brought into **two-part form** as required by Rule 43(1) EPC.

2 When amending the application, the applicant should make sure that no subject-matter is added **which extends beyond the application as originally filed** (Article 123(2) EPC) and he should indicate for each amendment from

which part of the application as originally filed said amendment can be derived directly and unambiguously. At the same time, he should ensure to avoid any inadmissible intermediate generalizations (Guidelines H-V 3.2.1).

3 The applicant should bring the **description** into **conformity with** the amended **claims**. The applicant should ensure to avoid inconsistencies between the description and claims (Guidelines F-IV 4.3). After amendments, all features of the independent claim must be indicated as mandatory throughout all instances in the description. Embodiments not falling under the scope of the claims must be deleted or clearly indicated as not forming part of the invention at all instances throughout the description. Specifically, the **following requirements** should be fulfilled by the description:

- The **technical field** of the invention (Rule 42(1)(a) EPC; Guidelines F-II. 4.2) must correspond to the amended set of claims.
- The **summary of the invention** (Rule 42(1)(c) EPC; Guidelines F-II.4.5) must correspond to the amended set of claims. Where appropriate, statements such as “The invention is set out in the appended set of claims“ or “The invention is as defined in claim X“ may be used instead of repeating the claims verbatim.
- **Embodiments** which are no longer covered by the claims must be deleted, unless they can reasonably be considered to be useful for highlighting specific aspects of the amended subject-matter. If this is the case, they must be clearly marked as not being part of the present invention (T 1808/06 and Guidelines F-IV, 4.3):
  - For example, ways of carrying out the invention using features presented as alternatives to features contained in the independent claims must either be deleted or the fact that they are not encompassed by the claims must be clearly stated.
  - Any combinations of features which do not fall under the scope of the amended independent claims must be clearly marked as not being part of the present invention.
  - Additionally, statements such as “The following examples/aspects/ embodiments X, Y, Z are not according to the invention and are present for illustration purposes only” may be used at the start of the description or in the list of figures.

– The **technical features of the independent claims** may **not** be presented as **optional** in the description. Where they precede a feature of an independent claim, terms such as **"for example"**, **"may"**, **"can"**, **"exemplary"**, **"optionally"**, **"preferably"** and the like must be removed.

If the above requirements with respect to the description are not met, the application can be refused under Article 97(2) EPC for reasons of lack of support by the description as required by Article 84 EPC.

The attention of the applicant is drawn to the fact that in the case of a further amended set of claims still giving rise to major objections with respect to patentability, **oral proceedings** are envisaged during which **refusal of the application is to be expected** if the objections are not overcome.